

The SmartTeach Manual

Integrating AI Tools in the AP CSP Curriculum

Welcome to the SmartTeach Manual, your comprehensive guide to harnessing the power of AI in the realm of education. In an era where technology is reshaping the way we teach and learn, this manual serves as your compass to navigate the exciting landscape of AI-enhanced education.

[Teaching and Learning Scenarios](#)

Explore example scenarios that illustrate how AI can be integrated seamlessly into various educational contexts. These scenarios are designed to spark inspiration and provide practical examples for teachers looking to enhance their teaching methodologies.

[Mini-Lessons and Activities](#)

Get started with starter mini-lessons and activities that leverage AI to make learning more dynamic, interactive, and effective. These resources are tailored to help you make the most of AI in your lessons.

[Using AI on the Create Task](#)

Investigate strategies for supporting students with using AI to develop their program for the Create Task.

Each section of this manual is carefully crafted to empower educators and learners alike with the knowledge and tools needed to thrive in the digital age. Whether you're seeking innovative teaching strategies, engaging activities, or insights into leveraging AI for educational success, you'll find it here.

Teaching and Learning Scenarios

This section is designed to provide examples of innovative ways AI can be woven into the curriculum. These are intended to provide inspiration for how you might use AI tools in different contexts and help you think creatively about how you might use them in your own classroom.

For a CSP-specific lesson outline using these tools, check out the [Mini-Lessons and Activities](#) section!

Fact or Fiction

 AI Tool: [The AI Detector](#)

A journalism class is discussing the impact of AI on news creation. The teacher introduces AI Detector Pro to explore how AI-generated content can blend with human writing.

Students are tasked with writing short news articles, some using AI assistance. They then use AI Detector Pro to analyze each other's articles, determining which ones might be AI-generated. This exercise helps students understand the capabilities of AI in content creation and the importance of critical analysis in journalism.

Digital Brushstrokes

 AI Tool: [Freeflo](#)

An art teacher wants to showcase the intersection of technology and art to her students. She decides to use Freeflo to demonstrate generative AI in creating art.

Students use Freeflo to generate images based on specific prompts related to their current art project. They then create physical art pieces inspired by these AI-generated images. This session aims to explore how AI can be used as a tool for artistic inspiration and creativity.

Creative Sparks

 AI Tool: [Adobe Firefly](#)

In a high school visual arts class, students are learning about graphic design. The teacher introduces Adobe Firefly as a tool to create unique graphics and textures.

Each student chooses a theme and uses Adobe Firefly to generate graphics and textures related to that theme. They then incorporate these AI-generated elements into a digital art project, learning about the blend of AI technology and artistic creativity.

Sounds of Speech

 AI Tool: [Riverside Audio Transcription](#)

In a language studies class, students are learning about different dialects and accents. The teacher uses Riverside Audio Transcription to transcribe various audio clips featuring diverse accents.

Students listen to the audio, read the transcriptions, and then discuss the nuances of spoken language and transcription accuracy. This activity helps them understand the challenges of audio transcription and the role of AI in overcoming language barriers.

Concept to Campaign

 AI Tool: [Catbird.ai](https://catbird.ai)

A group of marketing students is tasked with developing an advertising campaign for a new product. They decide to use Catbird for initial brainstorming and concept development.

Each student inputs prompts related to the product's features or target audience into Catbird to generate visual ideas. They use these images as inspiration to develop unique advertising concepts, learning how AI can be a valuable tool in the creative process of marketing and advertising.

Engineering the Future

 AI Tool: [Safurai](https://safurai.com)

In a university computer science course, students are learning about software development. The professor introduces Safurai to assist students in coding exercises.

Students use Safurai to write, optimize, and understand complex code segments. They engage with Safurai's interactive features to get explanations, optimizations, or unit tests for their code, learning about the practical application of AI in coding.

Mastering Literature with AI

 AI Tool: [PDF.ai](https://pdf.ai)

University researchers often deal with an overwhelming number of scholarly articles. To streamline their research process, they use PDF.ai.

Researchers upload multiple academic papers into PDF.ai and use its summarization feature to quickly grasp the main findings of each paper. This scenario demonstrates how AI can be utilized to efficiently process and summarize complex academic information, aiding in literature reviews and research.

Narrative Unleashed

 AI Tool: [Tome](https://tome.app)

In a high school literature class, the teacher introduces Tome as a tool for digital storytelling. Students are studying narrative techniques and are tasked with creating their own stories.

Each student uses Tome to generate the initial framework for their story based on given prompts. They further develop the story, learning how AI can aid in the creative process and understanding the mechanics of storytelling.

Mini-Lessons and Activities

The AI Detector

What's the tool?

[The AI Detector](#) by Content at Scale is a tool designed to identify whether text is generated by AI, including models like ChatGPT, GPT-4, Bard, and Claude. It claims to have a high accuracy rate of 98% and analyzes content for AI-generated characteristics. The tool also offers a pro version, which rewrites sentences and paraphrases paragraphs to make AI-generated content appear more humanlike, thus bypassing AI detection tools. This service is targeted towards various users, including marketers and content creators, to ensure their content maintains a human-like quality.

AP CSP Learning Objectives

- CRD-2.A.1: The purpose of computing innovations is to solve problems or pursue interests through creative expression.
- DAT-1.A.5: Abstraction is the process of reducing complexity by focusing on the main idea. By hiding details irrelevant to the question at hand and bringing together related and useful details, abstraction reduces complexity and allows one to focus on the idea.
- DAT-2.D.1: Programs can be used to process data to acquire information.
- DAT-2.E.1: Programs are used in an iterative and interactive way when processing information to allow users to gain insight and knowledge.

Materials Needed

- Computers with internet access
- Access to [The AI Detector](#)
- Sample texts (both AI-generated and human-written)
- Article: [She Was Falsely Accused of Cheating With AI — And She Won't Be the Last](#)

Introduction

Briefly introduce The AI Detector and its purpose.
Discuss the importance of distinguishing between AI-generated and human-written content.

Exploration

Have students use The AI Detector to analyze sample texts.
Discuss how complex processes like natural language processing (NLP) and semantic analysis are abstracted by the tool to provide a simple output (detecting AI-generated content) [[reference](#)].

Investigation

Examine how the tool processes and understands textual content. Use the tool to analyze different texts and extract information about their origins (AI-generated vs. human-created).

Demonstration

Explore how The AI Detector uses programs to process and analyze data. Discuss the underlying algorithms and their role in data analysis.

Hands-On Activity

Engage students in a hands-on activity where they use the tool to process various texts.

Group Discussion and Reflection

Facilitate a group discussion on the tool's effectiveness, its real-world applications, and the importance of evaluating marketing claims and the consequences of mistakes. Encourage students to reflect on how AI tools like this can impact content creation and verification.

Possible Extension or Homework

Assign students to find or create texts, use The AI Detector to analyze them, and write a report on their findings and understanding of the AI processes involved.

Freeflo

What's the tool?

[Freeflo](#) is a free service that provides reusable styles for AI image creation. Users can copy the prompt, change the subject, and then add it to their preferred image generator. Some image generators that support Freeflo styles include Midjourney, Stability, and Firefly.

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Materials Needed

- Computers with internet access
- Access to [Freeflo](#)

Introduction

Present Freeflo and its purpose in generative AI.
Explain generative AI and its role in creative contexts like image and text generation.

Exploration

Guide students to use Freeflo to create content using prompts.
Discuss how Freeflo abstracts complex AI processes and simplifies them for users.

Investigation

Engage students in a discussion on how prompts represent data in Freeflo.
Highlight how data (prompts) are used to generate creative AI content.

Application

Use Freeflo to transform prompts into images or texts.
Discuss the transformation of data (prompts) into information (generated content).

Demonstration

Explore how Freeflo's AI models process prompts to generate content. Discuss the underlying algorithms and their data processing methods.

Hands-On Activity

Conduct a hands-on activity where students use Freeflo. Emphasize the influence of user inputs on the AI's output.

Group Discussion and Reflection

Facilitate a group discussion on Freeflo's AI capabilities and real-world applications. Encourage students to reflect on the role of AI in creative processes.

Possible Extension or Homework

Assign students to create diverse content using Freeflo, reflecting on the AI processes involved.

Adobe Firefly

What's the tool?

[Adobe Firefly](#) is a creative AI tool developed by Adobe. It's designed to assist in creative processes by leveraging artificial intelligence. Firefly aims to simplify and enhance the workflow of creators, allowing them to focus more on creative aspects while the AI handles the more technical or repetitive tasks. The tool integrates with Adobe's ecosystem, providing a seamless experience for users of Adobe products.

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Materials Needed

- Computers with internet access
- Access to [Adobe Firefly](#)

Introduction

Present Adobe Firefly as a creative AI tool.
Explain its role in transforming inputs like text or sketches into graphic content.

Explore

Guide students to use Firefly to create graphics from inputs.
Discuss how Firefly abstracts these inputs into complex outputs.

Investigate

Engage students in a discussion on how inputs are represented and processed in Firefly.
Highlight the transformation of textual or sketch data into graphic content.

Apply

Use Firefly to convert basic inputs into graphics or textures.
Discuss the process of turning basic data into informative content.

Demonstrate

Explore how Firefly's AI models interpret data to generate content. Discuss underlying algorithms and data processing methods.

Hands-On Activity

Conduct a hands-on activity where students use Firefly. Focus on how user inputs dynamically influence AI output.

Group Discussion

Facilitate discussion on Firefly's capabilities and real-world applications. Encourage reflection on AI's role in creative processes.

**Possible Extension
or Homework**

Assign students to create a project using Firefly, reflecting on AI processes involved.

Riverside Audio Transcription

What's the tool?

[Riverside Audio Transcription](#) is a feature of Riverside.fm, a platform that provides high-quality recording and editing capabilities for podcasts and videos. It supports audio transcriptions in over 100 languages with 99% accuracy, enabling users to easily create captions and transcripts for their recordings. This AI-powered feature enhances the efficiency of content creation, particularly for podcasters, video creators, and marketers

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Materials Needed

- Computers with internet access
- Access to [Riverside Audio Transcription](#)

Introduction

Present the tool and its purpose in transcribing audio/video to text.
Discuss the role of AI in audio transcription.

Explore

Demonstrate how the tool converts audio data into text.

Apply

Transcribe a sample audio file using the tool.
Highlight how raw audio data is processed to extract text information.

Demonstrate

Explore how the tool's AI models process audio data.
Discuss the program's algorithms and methods in data processing.

Interactive Session

Engage students in an interactive session using the tool.
Focus on the dynamic nature of AI in processing and transcribing audio.

**Group Discussion
and Reflection**

Facilitate a discussion on the tool's effectiveness and its applications. Encourage students to reflect on AI's role in data processing.

**Possible Extension
or Homework**

Assign students to transcribe different audio files, analyzing the AI process.

Catbird

What's the tool?

[Catbird.ai](https://catbird.ai) is a free web app that uses AI to generate images from a single prompt. It can generate images from more than 15 models, including Stable Diffusion, Dreamlike Diffusion, and Openjourney. Catbird.ai is designed to be versatile and easy to use, and can benefit many people, including graphic designers, marketing professionals, content creators, and researchers. It uses advanced generative neural networks to create realistic and unique images based on user input.

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Materials Needed

- Computers with internet access
- Access to [Catbird.ai](https://catbird.ai)

Introduction

Present Catbird as an AI-driven image generator.
Discuss its role in transforming prompts into visual content.

Explore

Demonstrate how Catbird simplifies the image generation process.
Discuss abstraction in AI, focusing on user-friendly interfaces.

Apply

Have students use Catbird to generate images from various prompts.
Highlight the transformation of textual prompts into visual content.

Demonstrate

Explore Catbird's data processing capabilities.
Discuss how the AI algorithms interpret prompts to create images.

Hands-On Activity

Conduct a hands-on activity for students to experiment with different prompts.
Focus on the interactive nature of AI in visual content creation.

Group Discussion and Reflection

Facilitate discussion on Catbird's applications and AI's role in creative industries.
Encourage reflection on the process of transforming prompts into images.

Possible Extension or Homework

Assign students to create a series of images using Catbird, reflecting on the AI process.

PDF.ai

What's the tool?

[PDF.ai](#) is an AI-powered platform that can transform PDF documents into interactive files. PDF.ai can perform a variety of tasks, such as:

- Chatting: Users can chat with their PDFs, ask questions, get summaries, and find the information they need quickly and efficiently.
- Text recognition: AI-powered text recognition can convert scanned PDFs into editable text, making them easier to search, copy, and edit.
- Organization tools: PDF.ai has advanced search and organization tools.

PDF.ai can be used to efficiently manage legal agreements, financial reports, and other documents. It can also be great for students or for those studying new skills, theories, projects and more.

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Materials Needed

- Computers with internet access
- Access to [PDF.ai](#)

Introduction

Explain its purpose in transforming PDFs into interactive documents.
Discuss the role of AI in processing complex textual data.

Apply

Use PDF.ai to process and interpret data from sample PDFs.
Discuss how information is extracted and presented.

Demonstrate

Explore PDF.ai's data processing capabilities.
Discuss how the AI algorithms interpret and process document data.

Hands-On Activity

Conduct a hands-on activity for students to use PDF.ai.
Focus on the dynamic nature of data processing in AI systems.

**Group Discussion
and Reflection**

Facilitate discussion on PDF.ai's applications and effectiveness.
Encourage students to reflect on AI's role in document processing.

**Possible Extension
or Homework**

Assign students to process different documents using PDF.ai and report on their findings.

Safurai

What's the tool?

[Safurai](#) is an AI Code Assistant designed to enhance coding efficiency and quality. It offers features like a textbox for seeking information, advice, and code suggestions, and shortcuts for code explanations, optimizations, and unit tests. Safurai can be trained on specific projects for precise responses and supports natural language searches across projects. Compatible with several programming languages, it is free for individual use and offers tailored solutions for enterprises. Safurai is particularly useful for developers and software engineers, aiming to make coding faster, safer, and easier.

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Materials Needed

- Computers with internet access
- Access to [Safurai](#)

Introduction

Explain its functionality as an AI coding assistant.
Discuss AI's role in simplifying and enhancing programming tasks.

Explore

Demonstrate Safurai's capability to simplify complex programming tasks.
Discuss the concept of abstraction in programming aided by AI.

Demonstrate

Use Safurai in coding exercises to showcase AI-assisted code writing and refinement.
Discuss how AI processes programming language data.

Hands-On Activity

Engage students in a hands-on coding session using Safurai.
Highlight the dynamic, real-time data processing capabilities of AI in coding.

**Group Discussion
and Reflection**

Discuss Safurai's effectiveness and applications in software development.
Encourage students to reflect on the impact of AI in programming.

**Possible Extension
or Homework**

Assign coding tasks using Safurai, focusing on AI's role in the process.

Tome

What's the tool?

[Tome](#) combines an interactive canvas with an AI storytelling partner that can help users create presentations, outlines, or stories with one click. Users can record narration, set a theme, and choose a color palette and font. Tome's AI edit button can also be used to rewrite.

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Materials Needed

- Computers with internet access
- Access to [Tome](#)

Introduction

Explain its functionality as an AI storytelling tool.
Discuss AI's role in creative writing and narrative construction.

Explore

Demonstrate how Tome uses AI to enhance storytelling.
Discuss the innovation behind AI-driven content creation.

Demonstrate

Use Tome to create stories, focusing on program behavior and user interaction.
Discuss how AI influences the storytelling process.

Apply

Engage students in creating stories using Tome.
Highlight how the tool processes inputs and generates narrative content.

Group Discussion and Reflection

Discuss Tome's effectiveness and potential applications in creative fields.
Encourage students to reflect on the intersection of AI and creativity.

**Possible Extension
or Homework**

Assign a project to create a narrative using Tome, focusing on AI's role.

Glimpse Surf

What's the tool?

[Glimpse Surf](#) claims to redefine your browsing experience with the world's only "internet-enabled AI assistant". It transforms your web browser into a powerhouse of intelligence, combining conversational abilities, content generation, and real-time internet insights.

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Materials Needed

- Computers with internet access
- Access to [Glimpse Surf](#)

Introduction

Present its functionalities in AI-enhanced web browsing.
Discuss AI's role in understanding and interacting with web content.

Explore Computing Innovations

Demonstrate Glimpse Surf's features like content summarization.
Discuss the innovation behind AI in web browsing.

Program Behavior and Functionality

Use Glimpse Surf for browsing, focusing on its AI features.
Discuss how AI influences web content understanding and interaction.

Data Processing and Insight Generation

Engage students in using Glimpse Surf for research.
Highlight AI's role in processing and summarizing web content.

**Algorithm
Development and
Implementation**

Discuss the algorithms behind Glimpse Surf's AI capabilities.
Encourage students to consider how these algorithms process data.

**Group Discussion
and Reflection**

Discuss Glimpse Surf's effectiveness and potential in web browsing.
Reflect on the intersection of AI and web technology.

**Possible Extension
or Homework**

Assign a project to use Glimpse Surf for a research topic.

Using AI on the Create Task

Understanding AI Code Generators

In the landscape of computer science education, AI code generators stand as a testament to the rapid advancements in technology. To effectively incorporate these tools into the AP CSP curriculum, it is essential to first understand what they are and how they function.

What are AI code generators?

AI code generators are sophisticated software systems that use artificial intelligence to assist in writing, optimizing, and troubleshooting code. These tools have been trained on vast repositories of programming data, enabling them to suggest code snippets, complete coding tasks, and even debug existing code. They serve not just as assistants but also as learning aids, providing students with real-time examples of coding practices and solutions.

How do AI code generators work?

These generators function by analyzing the input provided by the user, such as a description of a task or a partially completed code segment. Utilizing their extensive training, they generate relevant code suggestions or solutions. This process involves complex algorithms that understand programming languages, coding syntax, and sometimes even the context or the objective of the given task.

Benefits in a Learning Environment

- AI code generators can offer multiple ways to solve a problem, exposing students to a variety of coding styles and techniques.
- They provide instant feedback and suggestions, allowing students to learn and correct their mistakes in real time.
- With the safety net of AI assistance, students might feel more encouraged to experiment and try complex coding tasks.
- These tools can help students overcome stumbling blocks in coding, making the learning process more efficient and less frustrating.

Limitations and Considerations

While AI code generators are powerful tools, it's important to recognize their limitations. They should not replace fundamental understanding or manual coding. There's a risk of students becoming overly reliant on these tools, which can hinder the development of problem-solving skills and a deep understanding of programming concepts. Therefore, it's crucial for you to guide students in using these tools judiciously, ensuring that they complement, rather than replace, traditional learning methods.

Guidelines for Using AI Code Generators in the Classroom

Here are some key guidelines to help you make the most of these innovative tools in a classroom setting:

1. **Set Clear Learning Objectives**

- Clearly define how and why AI code generators are being used. Ensure that their use aligns with the learning objectives of the course, particularly focusing on the AP CSP Create task.
- Communicate to students that these tools are meant to supplement their learning, not replace the need for understanding the fundamentals of coding and problem-solving.

2. **Foster Ethical Use**

- Emphasize the importance of academic honesty. Students should use AI tools for assistance, but their final submissions must be their original work, reflecting their understanding and effort.
- Teach students to acknowledge any assistance received from AI tools, fostering transparency and ethical use of technology.

3. **Encourage Critical Engagement**

- Encourage students to critically assess the code generated by AI. They should understand why a certain solution is proposed and be able to identify any limitations or errors in the AI-generated code.
- Use AI tools as a means to enhance problem-solving skills. Students should first attempt to solve problems independently before seeking AI assistance.

4. **Balanced Approach to Coding Tasks**

- Ensure students engage in manual coding to strengthen their coding skills and deepen their understanding of programming concepts.
- Utilize AI tools for complex or challenging tasks where students might benefit from additional guidance or for exploring different approaches to a problem.

Integrating AI Tools into the Create Task

Here are some strategies for effectively incorporating AI tools into the Create task:

1. **Understanding the Create Task**

- Begin by ensuring that students fully understand the requirements and objectives of the Create task. Clarify the expectations around complexity, creativity, and originality.
- Reinforce the key concepts and skills that the Create task aims to assess, such as algorithm development, abstraction, and programming proficiency.

2. **Role of AI in Idea Generation**

- Encourage students to use AI tools for brainstorming and exploring different ideas. AI can suggest various approaches to a problem, helping to spark creativity and innovation.
- Use AI to help students assess the feasibility of their ideas, providing insights into what's programmatically possible and guiding them in setting realistic goals.

3. **AI-Assisted Coding**

- Students can use AI tools to help draft initial versions of their code. This can be particularly useful for more complex or unfamiliar aspects of their project.
- AI suggestions can be used to refine and optimize code, ensuring efficiency and adherence to best practices.

4. Troubleshooting and Debugging

- AI tools can assist in identifying and explaining errors in code, providing an additional layer of support for students.
- Encourage students to use AI-generated suggestions as one of several strategies for debugging their code, emphasizing the importance of understanding the underlying issues.

5. Enhancing Code Quality

- Use AI tools to conduct preliminary code reviews, offering suggestions for improvement on aspects like readability, structure, and documentation.
- AI can introduce students to industry-standard best practices, which they can learn from and apply in their own coding.

6. Reflective Documentation

- Encourage students to document their use of AI tools, reflecting on how these tools influenced their approach, design decisions, and problem-solving strategies.
- This documentation can also serve as a means for students to articulate their learning journey, including how they overcame challenges and what they learned.

Monitoring and Assessment

Here are strategies for you to monitor and assess the integration of AI tools in the classroom:

1. Tracking Student Progress

- Conduct periodic check-ins with students to discuss their use of AI tools and progress on the Create task. This allows for early identification of any over-reliance on AI.
- Set clear milestones for the Create task and monitor students' progress towards these goals, ensuring that AI tools are used appropriately at each stage.

2. Evaluating Ethical Use

- Encourage students to be transparent about their use of AI tools. This includes acknowledging the assistance they received and differentiating between AI-generated code and their own work.
- Perform random checks to ensure that students are complying with ethical guidelines, such as not presenting AI-generated code as entirely their own work.

3. Assessing Understanding

- Ask students to explain the logic behind the code they submit, especially sections where AI assistance was utilized. This can help assess their understanding of the concepts.
- Conduct brief oral examinations where students discuss their project and the role AI played in it. This provides insight into their comprehension and application of learning.

4. Quality of Code

- Organize code review sessions where students review and provide feedback on each other's code. This encourages a deeper understanding and refinement of coding skills.
- Compare the quality of student code against established coding standards and best practices to assess proficiency and skill development.

5. Reflective Practice

- Have students maintain journals or logs where they reflect on their use of AI tools, the challenges faced, and how they overcame them. This promotes self-assessment and critical thinking.
- Provide feedback on these reflective entries to guide students' understanding and encourage deeper engagement with the learning process.